

S.M.N. RATHNAYAKE

Undergraduate — B.Sc. (Hons) Information Technology & Management

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ABOUT ME

IT undergraduate seeking a Software Engineering / Full-Stack Development internship. Full-stack software skills — independently architecting and shipping production web applications, including a multi-provider AI platform with RLS-secured backend logic, real-time features, and production deployment across Vercel and Railway. Comfortable owning a project end-to-end, from database schema design to live infrastructure; this end-to-end mindset extends to hardware, having designed a custom PCB and sensor-driven circuit for an embedded systems project.

EDUCATION

University of Moratuwa Mar 2024 – Present
B.Sc. (Hons) in Information Technology & Management — Faculty of Information Technology
GPA 3.67 / 4.00 through Semester 3 · Completed 2nd Year

National Institute of Business Management (NIBM) Mar 2023 – Jul 2023
Studied Computer Science

British Council Apr 2023 – Aug 2023
English Language & Public Speaking

G.C.E. Advanced Level (Physical Science Stream) — Combined Mathematics, Physics, Chemistry

PROJECTS

Digital Employee Performance Management System (Dart Global Logistics Inc.) 2026
Software Development Project — IS2901 | Client: Dart Global Logistics Inc. (Colombo) | Team of 5 | Python Flask, Next.js, TypeScript, Supabase (PostgreSQL), APScheduler
Repo: github.com/madhusiva19/pms (Contributor)

- Built a role-based Performance Management System for a 25-country aviation and logistics client, spanning a 5-tier role hierarchy from Global Admin down to Sub-Department Admin.
- Designed and built the HQ Admin (Global) Template Management module end-to-end — template creation, editing, assignment, freeze management, and variants applied worldwide across all countries and branches — plus history, appraisal-cycle configuration, and automated cut-off notifications.
- Developed the Template Dashboard with multi-dimensional filtering, a live appraisal-cycle timeline, and real-time freeze-state indicators across a 6-level organisational hierarchy.
- Built Flask REST endpoints and backend logic for template CRUD, variant management, freeze/unfreeze exceptions, and cycle rollover, with role-based access control and structured error handling.
- Designed the core database schema and configured 40+ real performance templates with 400+ assignment rules from client data.

LexaSpeakAI — Production AI English Coaching Platform 2026
Individual Project | Next.js 14, TypeScript, Express.js, Supabase (PostgreSQL), Google Gemini API, Anthropic Claude API, Tailwind CSS, Framer Motion, Vercel, Railway
Live: lexa-speak-ai.vercel.app | Repo: github.com/NethminiRathnayake-03/LexaSpeakAI

- Architected and shipped a production full-stack platform as two independently deployed services — Next.js 14 frontend (Vercel) and Express.js REST API backend (Railway) — sharing a Supabase database secured with RLS and server-side JWT authentication.

- Engineered a multi-provider LLM chain (Gemini -> Claude -> rule-based fallback) powering real-time grammar correction and AI-graded assessments, degrading gracefully under API failures and rate limits.
- Built a real-time AI conversation experience with Free Chat and scenario-based Roleplay across 10 real-world scenarios.
- Designed a CEFR-aligned 5-level × 10-lesson learning progression across 8 practice modes, with AI-graded advancement tests and per-mode score tracking.
- Built audio-driven practice via four accent- and gender-matched AI coaches using the Web Speech API.
- Implemented the full social and auth layer — friend requests, team management, RLS-scoped search, onboarding, profile management — and shipped installable Android/iOS builds.

Vantage — Immersive 3D Shopping Platform

2026

Individual Project | Next.js, React, TypeScript, Tailwind CSS, Framer Motion, Flask, PostgreSQL (Neon)

Live: frontend-kappa-sepia-51.vercel.app | Repo: github.com/NethminiRathnayake-03/Immersive-3D-Virtual-Marketplace

- Independently built a full-stack 3D e-commerce platform letting customers view products in real-time 3D, place them in AR via phone camera, and recolor them live before purchase.
- Built and deployed the full e-commerce flow — session-based auth, cart, checkout, order history — with Flask REST APIs backed by PostgreSQL.
- Designed a motion-driven, immersive UI (custom cursor, scroll-triggered animations, magnetic buttons, animated page transitions).
- Integrated Google's model-viewer library for real-time 3D rendering, AR placement, and live material/color editing.
- Deployed end-to-end to production (Vercel + Neon PostgreSQL), resolving cross-domain authentication and CORS in a live environment.

Smart Abacus (AbacusGenius) — Microcontroller-Based ICT Project

2025

IS1901 Level 01 Project | Team of 5 | Arduino Mega, Hall Effect sensors, ESP8266, DFPlayer Mini, HTML/CSS/JS/MQTT | Selected for FITE expo 2025

- Co-developed an interactive electric abacus pairing physical bead movement with real-time digital feedback for teaching arithmetic to children.
- Designed a Hall Effect sensor-based bead-counting mechanism, replacing an unreliable magnet/iron-ball design.
- Owned the audio-visual/control subsystem end-to-end: DFPlayer Mini audio, LED feedback, and button input.
- Diagnosed and fixed a short-circuit fault on the custom PCB that had disabled the Arduino board.
- Built a companion web app communicating with the device over MQTT via ESP8266.

Inkwell (MyBlog) — Personal Full-Stack Blogging Platform

2026

Individual Project | PHP, MySQL, HTML/CSS/JS

Live: nethzblog.42web.io | Repo: github.com/NethminiRathnayake-03/InkWell

- Independently built and deployed a full-stack blogging platform with auth and full CRUD, then ran a full security, feature, and UI overhaul post-launch.
- Hardened the app against common vulnerabilities: stored-XSS fixes, CSRF-protected POST for destructive actions, rate-limited login attempts, re-encoded uploads.
- Built comments/likes with AJAX, cover-image uploads, categories/tags, paginated search, and a redesigned dark-mode UI.
- Deployed and managed the app on live infrastructure end-to-end, diagnosing real hosting issues.

COMPETITIONS & EXTRACURRICULAR ACTIVITIES

SPARK Competition — University of Moratuwa

- Participated in an innovation-focused competition organised by the Department of Electronic and Telecommunication Engineering.

CodeRush Competition — University of Moratuwa

- Engaged in a coding competition emphasising algorithmic thinking and teamwork.

- Technical writing: publish in-depth write-ups on personal engineering projects covering architecture, trade-offs, and debugging.

Community & Club Involvement

- Member, Rotaract Club
- Member, Leo Club
- Active Member of the School Swimming Team; Selected for the University Swimming Team

KEY SKILLS

Programming Languages: Python, PHP, Java, C, JavaScript, TypeScript, SQL

Web & Backend: Flask, Next.js, React, Node.js, REST APIs, Supabase (PostgreSQL), MySQL, APScheduler

Mobile App Development: Packaging and shipping Next.js/React web platforms as installable cross-platform mobile apps for Android & iOS, responsive mobile-first UI design

Embedded & Hardware: Arduino Mega, circuit debugging

Tools & Practices: Git/GitHub, Agile Scrum, system/database design, phpMyAdmin, InfinityFree/Vercel/Railway deployment, structured debugging & troubleshooting

Core Strengths: Backend architecture & automated scheduling systems, cross-functional teamwork, hardware-software integration, technical writing

Communication: Fluent in English (written & spoken), public speaking; experienced presenting to and gathering requirements from external client stakeholders

AREAS OF INTEREST

Software engineering · Software development · Full-stack development · UI/UX design · Backend development · Mobile app development · Quality assurance · Business analysis · Project management · Embedded/IoT systems · Educational technology